D07 · Warehouse Automation & AMR/Robotics

05 · Automation ROI, total cost of ownership & implementation roadmap

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

The automation business case compares the total cost of ownership of automated options with continuing manual operation over the same horizon, and sequences investments in a roadmap from lower to higher capital and risk.

2 WHY IT MATTERS

Automation proposals often show capex and labor savings, and leave out integration, maintenance, software, specialized staff and downtime. **An honest TCO, built on the site's own data and compared with a do-nothing scenario, is what separates an investment from a bet.** Phasing lets the company learn from lower-risk steps before committing major capital.

3 HOW IT IS DONE

1. **Use your own operating data.** Volumes, productivity and costs from the site, not generic benchmarks.
2. **Build the full TCO.** Capex, integration, maintenance, software, staff, training and downtime over the horizon.
3. **Compare against do-nothing.** Include projected labor cost and volume growth in the manual baseline.
4. **Phase the roadmap with decision gates.** Lower-risk steps first; later phases conditional on volume thresholds.

IN PRACTICE

Evaluate the full TCO — not just automation capex

Annual maintenance at 10% of capex, accumulated over 5 years, is 50% of the original capex. Omitting it underestimates the real TCO and generates incorrect investment decisions.

Implement automation in phases — from lowest to highest cost and risk

Starting with the lowest-capex, highest-ROI technologies (voice-directed, AMRs) before higher-capex ones (ASRS, sorters) allows learning with real data before committing major capital.

Use real operation productivity and cost data in the business case

A business case based on industry benchmarks instead of real data can have ±30–50% errors in projected savings.

The automation business case must include the “do nothing” scenario

The base scenario must compare automation TCO vs. manual operation TCO over the same period — including projected labor cost growth.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Automation roadmap — distribution DC, 3 phases over 5 years — Evaluation of a 3-phase, 5-year automation investment roadmap.

Phase and technology	Capex	Projected ROI
Phase 1 (Year 1): Voice-directed picking + RF scanners in all zones	$180K	Pick accuracy: +0.7 pp · LPPH: +24% · Payback: 14 months
Phase 2 (Year 2): 12 collaborative AMRs for top 40% of picking volume	$840K	LPPH: +80% in AMR zones · -18 operators · Payback: 22 months

Phase and technology	Capex	Projected ROI
Phase 3 (Year 4–5): Pallet ASRS for bulk zone (if volume justifies)	$3.2M	Bulk DC DIO: -30% · Cost per pallet: -42% · Payback: 36 months

Result: Roadmap approved. Phase 1 generates the fastest ROI (payback 14 months) with the least risk. Phase 3 is conditioned on Year 3 volume exceeding 1,200 pallets/day — the threshold that justifies the ASRS.

HOW IT IS MEASURED

Automation Investment ROI %

Automation Investment ROI %(Total savings generated by automation / Total cost [capex + opex]) × 100

Benchmark: >200% ROI in 5 years to justify a warehouse automation project An automation project with <150% ROI in 5 years is high risk — payback is >3.3 years and assumes labor cost and volume will remain stable throughout that period.

Automation Payback Period (months to capex recovery)

Automation Payback Period(Automation capex / Monthly savings generated by automation)

Benchmark: <24 months for low-risk projects (AMRs, voice-directed) · <36 months for high-investment projects (ASRS, sorters) Payback period is the most commonly used indicator by CFOs to approve automation projects.

WHAT YOU WOULD USE

Automation Consulting

Gartner Magic Quadrant for Warehouse Management Systems — Module: Automation Market IntelligenceGartner publishes a Magic Quadrant for Warehouse Management Systems and market guides on warehouse automation — useful as a starting point for a vendor long list, not as a decision.

Interact Analysis / LogisticsIQ — Module: Warehouse Automation AnalyticsSpecialist analytics firms in the automation market with ROI benchmarks by technology.

4 SO WHAT

Decide automation on full TCO and phase it. Use your own data, count every cost, compare with the manual future and tie later phases to volume gates.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México